CALCULUS · LIMITS AND CONTINUITY

Limit Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Limit Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Let $f(x) = \frac{(x^2 - 1)}{(x - 1)}$. What is f(1)? What do you think f(x) is close to when x is close to 1?

   f(1) is undefined ($\frac{0}{0}$). f(x) is close to 2.

2. Can a function have a limit at a point where it is not defined?

   Yes. The limit only depends on values near the point.

2 Explore use the tool

1. Hole: record f(0.99) and f(1.01). What number do both sides approach?

   1.99 and 2.01; both approach 2.

2. Moved point: set f(2) to three different values. Does the limit change?

   No. The limit stays 3 whatever f(2) is.

3. Jump: with h = 0.001, record the left and right values. Then reveal the limits.

   1.999 and 0.001. Left limit 2, right limit 0, so the two-sided limit does not exist.

4. Oscillation: zoom in four times. Does the graph settle near one height?

   No. It keeps oscillating between −1 and 1, so the limit does not exist. Zooming alone can't prove a limit either way.

3 Record use the tool and fill in the table

Limits and values

Reveal the limits after you make a conjecture.

Function	Left limit	Right limit	Limit	f(a)	Continuous?
Hole, a = 1	2	2	2	undefined	no
Moved point, a = 2, f(2) = 5	3	3	3	5	no
Jump, a = 1	2	0	does not exist	0	no
Vertical asymptote, a = 2	∞	∞	∞ (no real limit)	undefined	no
Continuous, a = 2	1	1	1	1	yes

4 Explain what did you discover?

1. $\lim_{x \to a} f(x)$ describes the values of f ___near___ a, not ___at___ a.

2. The two-sided limit exists only when the ___left-hand___ limit and the ___right-hand___ limit are the same number.

3. f is continuous at a when f(a) is ___defined___ and equals ___the limit___.

4. A table shows $f(0.1) = 0$, $f(0.01) = 0$ and $f(0.001) = 0$. Can you conclude that the limit as $x \to 0$ is 0? Explain with an example.

   No. A table only checks some x-values. For example, $f(x) = \sin(\frac{\pi}{x})$ is 0 at 0.1, 0.01 and 0.001, but it oscillates between –1 and 1 between those values, so its limit at 0 does not exist.

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5 Apply a new situation, no tool

1. Find $\lim_{x \to 3} \frac{(x^2 - 9)}{(x - 3)}$.

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2. $g(x) = x + 1$ for $x < 2$ and $2x - 1$ for $x \geq 2$. Does $\lim_{x \to 2} g(x)$ exist? Is g continuous at 2?

   Left limit 3, right limit 3, so the limit is 3. $g(2) = 3$, so g is continuous at 2.

3. Describe the behavior of $h(x) = \frac{(x + 1)}{(x - 4)}$ as $x \to 4$.

   Vertical asymptote: from the left $h \to -\infty$, from the right $h \to +\infty$. No limit.

4. $f(x) = \frac{(x^2 - 1)}{(x - 1)}$ for $x \neq 1$ and $f(1) = k$. Find k so that f is continuous at 1.

   $k = 2$

Teacher notes

Intended discovery: A limit describes values near a, not at a: it exists only when both one-sided limits agree, f(a) can differ from it or be undefined, and tables or zooming are evidence, not proof.

Likely misconceptions:

- The limit is f(a), or there is no limit when f(a) is undefined.
- A table or a zoomed graph proves the limit (the Explain question gives a counterexample).
- Writing "$= \infty$" as if the limit exists as a number.

Answers that can vary: Moved-point values students choose in Explore 2; the description of the oscillation.